

Stochastic programming

Lagrangian methods

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Motivation

Reference: Birge and Louveaux, Sections 3.4, 5.8 and 6.4.

We consider the general **two-stage nonlinear program**

$$\begin{aligned} \inf z &= f^1(x) + Q(x) \\ \text{s.t. } g_i^1(x) &\leq 0, \quad i = 1, \dots, m_1, \end{aligned}$$

where $Q(x) = E_\xi[Q(x, \xi)]$, and

$$\begin{aligned} Q(x, \xi) &= \inf f^2(y(\xi), \xi) \\ \text{s.t. } b_i^2(x, \xi) + g_i^2(y(\xi), \xi) &\leq 0, \quad i = 1, \dots, m_2. \end{aligned}$$

We assume that all functions $f^2(\cdot, \xi)$, $b_i^2(\cdot, \xi)$ et $g_i^2(\cdot, \xi)$ are continuous in the first argument for any given ξ , and measurable in ξ for any fixed first argument. ξ has finite second-order moments.

Properties - assumptions

We can extend the definitions of K_1 and K_2 .

$$K_1 = \{x \mid g_i^1(x) \leq 0, i = 1, \dots, m_1\},$$

$$K_2(\xi) = \{x \mid \exists y(\xi) \text{ s.t. } b_i^2(x, \xi) + g_i^2(y(\xi), \xi) \leq 0, i = 1, \dots, m_2\},$$

$$K_2 = \{x \mid Q(x) < \infty\}.$$

Assumptions.

1. **Convexity.** The function f^1 is convex on $\mathbb{R}^{n_1}$, g_i^1 is convex on $\mathbb{R}^{n_1}$ ($i = 1, \dots, m_1$), $f^2(\cdot, \xi)$ is convex on $\mathbb{R}^{n_2}$ for all $\xi \in \Xi$, $g_i^2(\cdot, \xi)$ is convex on $\mathbb{R}^{n_2}$ ($i = 1, \dots, m_2$), for all $\xi \in \Xi$, and $b_i^2(\cdot, \xi)$ is convex on $\mathbb{R}^{n_1}$ ($i = 1, \dots, m_2$), for all $\xi \in \Xi$.
2. **Slater condition** If $Q(x) < \infty$, for almost every (a.e.) $\xi \in \Xi$, $\exists$ some $y(\xi)$ s.t. $b_i^2(x, \xi) + g_i^2(y(\xi), \xi) < 0$ ($i = 1, \dots, m_2$).

Second-stage properties

The Slater condition ensures that the *strong duality* holds at the second stage and the KKT conditions are necessary and sufficient.

Theorem

Under assumptions 1 and 2, the recourse function $Q(x, \xi)$ is a convex function in x for all $\xi \in \Xi$.

Theorem

If the recourse feasible set (i.e. in y , for given ξ) is bounded for any $x \in \mathbb{R}^{n_1}$, then the function $Q(x, \xi)$ is lower semi-continuous at x for all $\xi \in \Xi$.

Lower semi-continuity

We say that a function f is **lower semi-continuous at x_0** if for every $\varepsilon > 0$, $\exists$ a neighborhood U of x_0 such that $f(x) > f(x_0) - \varepsilon$ for all $x \in U$. Equivalently

$$\liminf_{x \rightarrow x_0} f(x) \geq f(x_0),$$

where

$$\liminf_{x \rightarrow a} f(x) = \lim_{\varepsilon \rightarrow 0} \inf \{ f(x) : x \in \text{dom}(f) \cap B(a; \varepsilon) \setminus \{a\} \}.$$

The function f is **lower semi-continuous** if it is lower semi-continuous at any point in its domain.

Lower semi-continuity

A function is lower semi-continuous iff $\{x \in X : f(x) > \alpha\}$ is an open set for every $\alpha \in \mathbb{R}$, or, in a similar way, $\{x \in X : f(x) \leq \alpha\}$ is a closed set for every $\alpha \in \mathbb{R}$.

Example of lower semi-continuous function

$$f(x) = \begin{cases} x & \text{if } x \leq 1 \\ x + 1 & \text{if } x > 1 \end{cases}$$

Convexity (cont'd)

Corollary

The expected recourse function $Q(x)$ is a convex function in x .

Corollary

The feasible set $K_2 = \{x \mid Q(x) < \infty\}$ is closed and convex.

Corollary

If the recourse feasible set is bounded for any $x \in \mathbb{R}^{n_1}$, Q is a lower semi-continuous function.

Solution: existence

Theorem

If the recourse feasible set is bounded for any $x \in \mathbb{R}^{n_1}$, K_1 is bounded, f^1 is continuous, g_i^1 and g_i^2 are continuous for every i , and $K_1 \cap K_2 \neq \emptyset$, then the nonlinear stochastic two-stage program has an optimal solution and the infimum is reached.

Solution: optimality

Theorem

Assume that the Slater condition is satisfied:

$$\exists x \text{ s.t. } x \in \text{ri}(\text{dom}(f^1(x))), x \in \text{ri}(\text{dom}(\mathcal{Q}(x))) \text{ and } g_i^1(x) < 0, i = 1, \dots, m_1.$$

x^ is optimal iff there exists λ^* such that (x^*, λ^*) satisfies the KKT conditions for the two-stage stochastic program, i.e.*

- $x^* \in K_1$,
- $\lambda^* \geq 0$,
- $\lambda_i^* g_i^1(x^*) = 0, i = 1, \dots, m$,
- $0 \in \partial f^1(x^*) + \partial \mathcal{Q}(x^*) + \sum_{i=1}^{m_1} \lambda_i^* \partial g_i^1(x^*)$.

Relative interior

The **relative interior** of a set S , denoted by $ri(S)$, is defined as its interior within the affine hull of S . In other words

$$ri(S) = \{x \in S \mid \exists \varepsilon > 0, (B_\varepsilon(x) \cap \text{aff}(S)) \subseteq S\},$$

where $\text{aff}(S)$ is the **affine hull** of S , and $B_\varepsilon(x)$ is a ball of radius ε centered at x . Any metric can be used for the ball construction: all define the same relative interior.

The **affine hull** $\text{aff}(S)$ of S is the set of all affine combinations of elements of S , i.e.

$$\text{aff}(S) = \left\{ \sum_{i=1}^k \alpha_i x_i \mid x_i \in S, \alpha_i \in \mathbb{R}, \sum_{i=1}^k \alpha_i = 1, k = 1, 2, \dots \right\}.$$

Basic principle of Lagrangian approach

- Nonlinear problem.
- We aim to build a search direction and compute a step along this direction to reduce the objective.
- Problem: classical nonlinear methods assume that the (sub-)gradients of $\mathcal{Q}$ are available and cheap to obtain. Not the case here.
- Link the first and second stages in order to avoid optimization subproblems when building search directions.

Basic principle of Lagrangian approach (cont'd)

- Let π be a vector of (dual) multipliers associated to the second-stage constraints.
- Dual problem with respect to the second-stage constraints (linking the first and second stages):

$$\max_{\pi(\xi) \geq 0} w = \theta(\pi),$$

where

$$\begin{aligned} \theta(\pi) = & \inf_{x, y} f^1(x) + \mathbb{E}_{\xi}[f^2(y(\xi), \xi)] + \\ & \mathbb{E}_{\xi} \left[\sum_{i=1}^{m_2} \pi_i(\xi) \left(b_i^2(x, \xi) + g_i^2(y(\xi), \xi) \right) \right] \\ \text{s.t. } & g_i(x) \leq 0, i = 1, \dots, m_1. \end{aligned}$$

Duality

Theorem

Assume that

- *all the functions of the stochastic program are convex,*
- *there exists a finite optimal value,*
- *the Slater constraints qualification condition holds,*
- *the support of ξ is finite.*

Then $z \geq w$ for all $x, y_1, \dots, y_S$ feasible in the primal formulation and $\pi_1, \dots, \pi_S$ feasible in the dual formulation (weak duality).

Moreover, their optimal values coincide: $z^ = w^*$ (strong duality).*

We thus assume from now that Ξ is finite, and we will describe some procedures exploiting the Lagrangian function.

Basic Lagrangian dual ascent method

Assumption: the dual problem always has a unique solution.

Algorithm 1: Lagrangian dual ascent method

Step 0. Let $\pi^{0,s} \geq 0$, $s = 1, \dots, S$, $\pi^0 = (\pi^{0,1}, \dots, \pi^{0,S})$, $\nu = 0$.

Step 1. Given $\pi = \pi^\nu$ in the dual problem, consider the solution $(x^\nu, y_1^\nu, \dots, y_S^\nu)$. For $s = 1, \dots, S$, define

$$\hat{\pi}_i^s = b_i^2(x^\nu, s) + g_i^2(y_s^\nu, s).$$

If $\hat{\pi}^s = 0$ for all s , stop.

Step 2. Choose a step $\alpha^\nu \geq 0$ and set $\pi^{\nu+1,s} = \max\{\pi^{\nu,s} + \alpha^\nu \hat{\pi}, 0\}$. Set $\nu := \nu + 1$ and check convergence. If we have not yet converged, return to Step 1.

Properties - convergence

- Various strategies can be used to compute the step α^ν . Assume we compute the step to maximize $\theta(\pi^\nu + \alpha \hat{\pi})$ on $\alpha \geq 0$. Under the assumption of uniqueness of dual solution, we can show that this algorithm always produces an **ascent direction** in θ .
- Either the algorithm converges to an optimal solution, or, assuming a bounded set of optimal solutions, it produces an infinite sequence where all limit points are optimal.
- Good performance if the number of dual iterations is small compared to the number of function evaluations that would be required by solving the original problem directly.
- Solving the dual program may be time-consuming, but should nevertheless be easier than solving the original problem directly, as the constraints linking the two stages (i.e. involving x and $y(\xi)$) now appear in the objective.

Another approach: scenarios

As Ξ is finite, we can consider that S scenarios exist, and we can link a non-anticipative decision $\hat{x}$ and the decisions of the scenario s with the terms $(\hat{x} - x_k)$ in the objective function.

Consider the general problem

$$\inf_{x \in \mathcal{N}} \mathbb{E} \left[\sum_{t=0}^T f_{t+1}(\xi, x^t(\xi), x_{t+1}(\xi)) \right],$$

s.t. $x_t(\xi) \in X_t(\xi)$ a.s.,

where x_t is the decision at stage t , and x^t is the decisions history at stage t , i.e. $x^t = \{x_1, \dots, x_t\}$. $\mathcal{N}$ designs the closed linear subspace of nonanticipative processes. Denote $\mathcal{A}$ the σ -field of all the events (i.e. the collection of events subsets, with the probability measure associated to ξ).

Nonanticipativity

Mathematically,

- the nonanticipativity aims to require that $x^t(\xi)$ has to be $\mathcal{A}^t$ -measurable, where $\mathcal{A}^t$ is the σ -field of events until time t ;
- or, in other words, $x^t(\xi) = E[x^t(\xi) | \mathcal{A}^t]$ a.s., $t = 0, \dots$;
- using the projection operator $\Pi^t : z \rightarrow \Pi^t z := E[z | \mathcal{A}^t]$, this is equivalent to

$$(I - \Pi^t)x^T = 0, \quad t = 0, \dots$$

Nonanticipativity (cont'd)

Let S_s^t be the set of scenarios identical to the scenario s at time t . We then have to ensure that

$$x_{its} = x_{its'}, \forall i \in N, \forall t \in T, \forall s \in S, \forall s' \in S_s^t.$$

Idea: explicitly include the nonanticipativity constraints, and place them in the objective function, rather than in the second stage constraints.

Dual ascent and scenarios

Consider the maximization of the dual built on the nonanticipativity constraints:

$$\max_{\pi(\xi) \geq 0} W = \theta(\pi),$$

where

$$\begin{aligned} \theta(\pi) = \inf_{x \in \mathcal{X}} z = & \mathbb{E} \left[\sum_{t=0}^T f_{t+1}(\xi, x^t(\xi), x_{t+1}(\xi)) \right] \\ & + \mathbb{E} \left[\sum_{t=0}^T \pi^t(\xi) (I - \Pi_t) x^t(\xi) \right], \end{aligned}$$

where π^t corresponds to the components of π associated to the t first periods.

Dual ascent and scenarios: two stages

We still have to describe the projection operator. For simplicity, consider two steps. The primal problem becomes (omitting all the constraints except the nonanticipativity ones)

$$\begin{aligned} \min z &= \sum_{s=1}^S p_s (f^1(x_s) + f^2(x_s, y_s)) \\ \text{s.t. } x_s - \sum_{k=1}^S p_k x_k &= 0, \quad s = 1, \dots, S. \end{aligned}$$

The dual problem can now be written as

$$\begin{aligned} \max_{\pi} \theta(\pi) = \min_{x, y} \sum_{s=1}^S p_s \bigg(& f^1(x_s) + f^2(x_s, y_s) \\ & + \pi_s \left(x_s - \sum_{k=1}^S p_k x_k \right) \bigg). \end{aligned}$$

Dual ascent and scenarios: algorithm

Algorithm 2: Lagrangian dual ascent method

- Step 0.** Let $\pi^0 \geq 0$, $\nu = 0$.
- Step 1.** Given $\pi = \pi^\nu$ in the dual problem, compute the solution $(x_1^\nu, \dots, x_S^\nu, y_1^\nu, \dots, y_S^\nu)$.
- Step 2.** If $x_s - \sum_{k=1}^S p_k x_k = 0$, $s = 1, \dots, S$, stop: the solution is optimal. Otherwise, define $\hat{\pi}_s = x_s - \sum_{k=1}^S p_k x_k$, and go to Step 3.
- Step 3.** Let α^ν maximize $\theta(\pi^\nu + \alpha \hat{\pi})$ on $\pi^\nu + \alpha \hat{\pi} \geq 0$, $\alpha \geq 0$. Set $\pi^{\nu+1} = \pi^\nu + \alpha^\nu \hat{\pi}$, $\nu = \nu + 1$. Return to Step 1.

Augmented Lagrangian

- Unfortunately, usually slow as there is a linearization at a single point of θ only. It is nevertheless easy to implement and can give good results, especially for small size problems.
- In order to improve the performances, we will turn to another technique inspired by nonlinear programming: **augmented Lagrangian** approaches.
- Augmented Lagrangian: add a penalty on $\theta(\pi)$.
- Moreover, with augmented Lagrangian techniques, we can try to exploit the problem structure in order to decompose the problem.

Augmented Lagrangian and Stochastic Programming

We again develop the ideas in the two-stages context, while the approach can be easily generalized in the multi-stage context.

Recall that we aim to solve

$$\max_{\pi(\xi) \geq 0} w = \theta(\pi),$$

where

$$\begin{aligned} \theta(\pi) = \inf_{x \in \mathcal{X}} z = & \mathbb{E} \left[\sum_{t=0}^T f_{t+1}(\xi, x^t(\xi), x_{t+1}(\xi)) \right] \\ & + \mathbb{E} \left[\sum_{t=0}^T \pi^t(\xi) (I - \Pi_t) x^t(\xi) \right], \end{aligned}$$

Penalizing the nonanticipativity constraints, we obtain the following program, where $\hat{x}$ is nonanticipative.

Augmented Lagrangian and SP: two stages

$$\begin{aligned} \theta(\pi) = \inf z = f^1(\hat{x}) + \sum_{s=1}^S \left(p_s f^2(y_s, \xi_s) + \pi_s^T (x_s - \hat{x}) + \frac{\rho}{2} \|x_s - \hat{x}\|^2 \right) \\ \text{s.t. } g_i^1(\hat{x}) \leq 0, \quad i = 1, \dots, m_1, \\ t_i^2(x_s, \xi_s) + g_i^2(y_s, \xi_s) \leq 0, \quad i = 1, \dots, m_2, \quad s = 1, \dots, S. \end{aligned}$$

ρ is the augmented Lagrangian penalty parameter, and π_s is the vector of dual variables associated to the nonanticipativity constraints of the scenario s .

The method aims to contract the pair $(\hat{x}^{\nu+1}, \pi^{\nu+1})$ around a saddle point.

Progressive hedging

- Main reference: R. T. Rockafellar and R. J.-B. Wets, *Scenarios and policy aggregation in optimization under uncertainty*, Mathematics of Operations Research 16(1):119–147 (1991).
The paper focussed on the multistage version.
- The method performs a complete separation of the problems, scenario by scenario, at each iteration, reducing the cost per iteration.
- The number of iterations can however increase. . .
- But for structured problems, it is possible to exploit parallelism, and therefore solve large-scale problems.
- **Observation:** the Lagrangian function is not separable with respect to the scenarios due to the term $(\hat{x} - x_s)$.

Progressive hedging: basic principle

Alternatively,

1. fix $\hat{x}$ and obtain the solutions x_s , $s = 1, \dots, S$,
2. fix x_s , $s = 1, \dots, S$, and compute $\hat{x}$.

In other words, we work scenario by scenario, and we enforce progressively the nonanticipativity constraints.

At iteration ν , we solve the subproblems

$$\begin{aligned} \inf z = & \sum_{s=1}^S p_s \left(f^1(x_s) + f^2(y_s, \xi_s) + (\pi_s^\nu)^T (x_s - \hat{x}^\nu) + \frac{\rho}{2} \|x_s - \hat{x}^\nu\|^2 \right) \\ \text{s.t. } & g_i^1(x_s) \leq 0, \quad i = 1, \dots, m_1, \quad s = 1, \dots, S, \\ & t_i^2(x_s, k) + g_i^2(y_s, \xi_s) \leq 0, \quad i = 1, \dots, m_2, \quad s = 1, \dots, S. \end{aligned}$$

$\hat{x}^\nu$ is not necessarily a feasible solution!

Decomposition

However, it is easy to decompose the problem!

Algorithm:

Step 0. Assume that we have a nonanticipative solution x^0 , an initial vector of multipliers π^0 , and $\rho > 0$. $\nu \leftarrow 0$. Go to Step 1.

Step 1. Let $(x_s^{\nu+1}, y_s^{\nu+1})$ be a solution of the previous Lagrangian program, for $s = 1, \dots, S$. Set

$$\hat{x}^{\nu+1} = \left(\hat{x}_1^{\nu+1}, \dots, \hat{x}_S^{\nu+1} \right)^T,$$

$$\text{where } \hat{x}_s^{\nu+1} = \sum_{s=1}^S \rho_s x_s^{\nu+1}, \quad s = 1, \dots, S.$$

Decomposition (cont'd)

Step 2. Let $\pi_s^{\nu+1} = \pi_s^\nu + \rho (x_s^{\nu+1} - \hat{x}_s^{\nu+1})$, $s = 1, \dots, S$.

If $\hat{x}^{\nu+1} = \hat{x}^\nu$ and $\pi^{\nu+1} = \pi^\nu$, stop: $(\hat{x}^\nu, \pi^\nu)$ is optimal.

Otherwise, set $\nu = \nu + 1$ and return to Step 1.

- Step 2 simply consists of taking the expected value of $x^{\nu+1}$ as $\hat{x}^{\nu+1}$.
- The basis of this approach is therefore the contraction of the pair $(\hat{x}^\nu, \pi^\nu)$ around a saddle point rather than a dual ascent strategy.
- In practice, the optimality test is replaced by a convergence test for instance if the following quantity is small enough (De Silva and Abramson):

$$\sqrt{\|\hat{x}^\nu - \hat{x}^{\nu+1}\|^2 + \sum_{s=1}^S \rho_s \|x_s^\nu - \hat{x}^\nu\|^2}.$$

Generalization to multistage

$$\begin{aligned} \min_x \quad & \sum_{s \in \mathcal{S}} p_s f(x^{(s)}, \xi^{(s)}) \\ \text{s.t.} \quad & x^{(s)} \in \mathcal{X}^{(s)}, \\ & x_t^{(s)} \text{ is nonanticipative, } t = 1, \dots, T. \end{aligned}$$

Nonanticipativity:

$$x_t^{(s)} = \mathbb{E} \left[x_t^{(s')} \mid s' \in \mathcal{S}_t^{(s)} \right],$$

where $\mathcal{S}_t^{(s)} = \{s' \mid \bar{\xi}_t^{(s)} = \bar{\xi}_t^{(s')}\}$. It can be reformulated as

$$x_t^{(s)} = \hat{x}_t^{(s)},$$

with

$$\hat{x}_t^{(s)} = \frac{\sum_{s' \in \mathcal{S}_t^{(s)}} p_{s'} x_t^{(s')}}{\sum_{s' \in \mathcal{S}_t^{(s)}} p_{s'}}.$$

Generalization

Augmented Lagrangian:

$$L(x, \pi, \rho) = \mathbb{E} \left[f \left(x^{(s)}, \xi^{(s)} \right) + \sum_{t=1}^T \left(\pi_t^{(s)'} \left(x_t^{(s)} - \hat{x}_t^{(s)} \right) + \frac{\rho}{2} \left\| x_t^{(s)} - \hat{x}_t^{(s)} \right\|^2 \right) \right],$$

where π is the Lagrange multipliers vector associated to the NA constraints, and $\rho > 0$ is a penalty parameter.

Implementation

The `StochasticPrograms.jl` package proposes an implementation of the progressive hedging algorithm, but limited to two stages.

Some variations or unmaintained versions exist.

Generalization

Given π , the augmented Lagrangian program is

$$\begin{aligned} \min_x \quad & L(x, \pi, \rho) \\ \text{s.t.} \quad & x^{(s)} \in \mathcal{X}^{(s)}, \mathbf{s} \in \mathcal{S}. \end{aligned} \tag{1}$$

In order to achieve full separability, fix $\hat{x}_t^{(s)}$ and repeatedly solve the program by updating the Lagrange multipliers vector and the value of $\hat{x}_t^{(s)}$ between consecutive resolutions.

Progressive hedging algorithm (PHA)

Step 0. Set $\hat{x}^{(s),0} = (\hat{x}_1^{(s),0}, \dots, \hat{x}_T^{(s),0})$ and $k = 0$. Choose $\pi^{(s),0} = \mathbf{0}$, $\rho^0 > 0$.

Step 1. Compute $x^{(s),k+1} = (x_1^{s,k+1}, \dots, x_T^{s,k+1})$, $s = 1, \dots, S$, by solving each scenario subproblem

$$\begin{aligned} \min_{x^{(s)}} & f(x^{(s)}, \xi^{(s)}) + \sum_{t=1}^T \left(\pi_t^{(s)'} \left(x_t^{(s)} - \hat{x}_t^{(s),k} \right) \right. \\ & \left. + \frac{\rho^k}{2} \left\| x_t^{(s)} - \hat{x}_t^{(s),k} \right\|^2 \right) \\ \text{s.t. } & x^{(s)} \in \mathcal{X}^{(s)}. \end{aligned}$$

Step 2. For $s = 1, \dots, S$, $t = 1, \dots, T$, set

$$\hat{x}_t^{(s),k+1} = \frac{\sum_{s' \in \mathcal{S}_t^{(s)}} \rho_{s'} x_t^{(s'),k+1}}{\sum_{s' \in \mathcal{S}_t^{(s)}} \rho_{s'}}.$$

Progressive hedging algorithm (PHA)

Step 3. Set ρ^{k+1} and

$$\pi_t^{(s),k+1} = \pi_t^{(s),k} + \rho^k \left(x_t^{(s),k+1} - \hat{x}_t^{(s),k+1} \right),$$

for $t = 1, \dots, T$, $s \in \mathcal{S}$.

Step 4. Stop if convergence is achieved. Otherwise, set $k \leftarrow k + 1$ and return to Step 1.

Convergence

Theorem

Assume that the initial nonlinear stochastic program has only convex functions and a finite optimal value, as well as a strictly feasible point. Assume moreover that the support of ξ is finite. The progressive hedging algorithm converges to an optimal solution x^, π^* .*

- De Silva and Abramson have tested the progressive hedging algorithm for linear problems in portfolio management, proposed by Mulvey and Vladimirou.
- The subproblems were solved by means of interior points methods.
- Good speed-up convergence, but the algorithm remains sensitive to the choice of the penalty parameter.